How-to guide

One task per section: when to use it, the steps, what changes and how to check it. Page names are those in the rail on the left (**1 Bring in**, **2 Organize**, **3 Correct**, **4 Report** and **More**); an orange number in a step points to the same marker in its figure. For a first project, start with the WLR quick start; for what WLR does to the data and why, see the WLR reference guide.

WLR 0.12.0 · 28 October 2026 · Figures: constructed example, not field data

BRING IN

1 Add a new download
2 Leave out a file or say what logger it is

ORGANIZE

3 Give a logger a site, dates or a hang length
4 Accept suggestions and decide what the run found
5 Rename a site or hide it
6 Correct the workbook for one deployment

CORRECT

7 Set the time zone of files that state none
8 Fix a logger clock
9 Choose the barologgers
10 Enter a logger offset or a survey point
11 Enter tape checks and check drift
12 Remove a stretch of data
13 Change a quality limit
14 Switch an automatic correction off

CHECK AND REPORT

15 Look at a site over time
16 Find out why a value was removed
17 Run the set-up check
18 Export data and reports
19 Back up or share the set-up
20 Report a problem

The three views

The same project, data and edits in three views, switched at the top (**Simple**, **Standard**, **Workbench**) or with ?ui=simple, ?ui=standard or ?ui=workbench in the address. A new install opens in Simple.

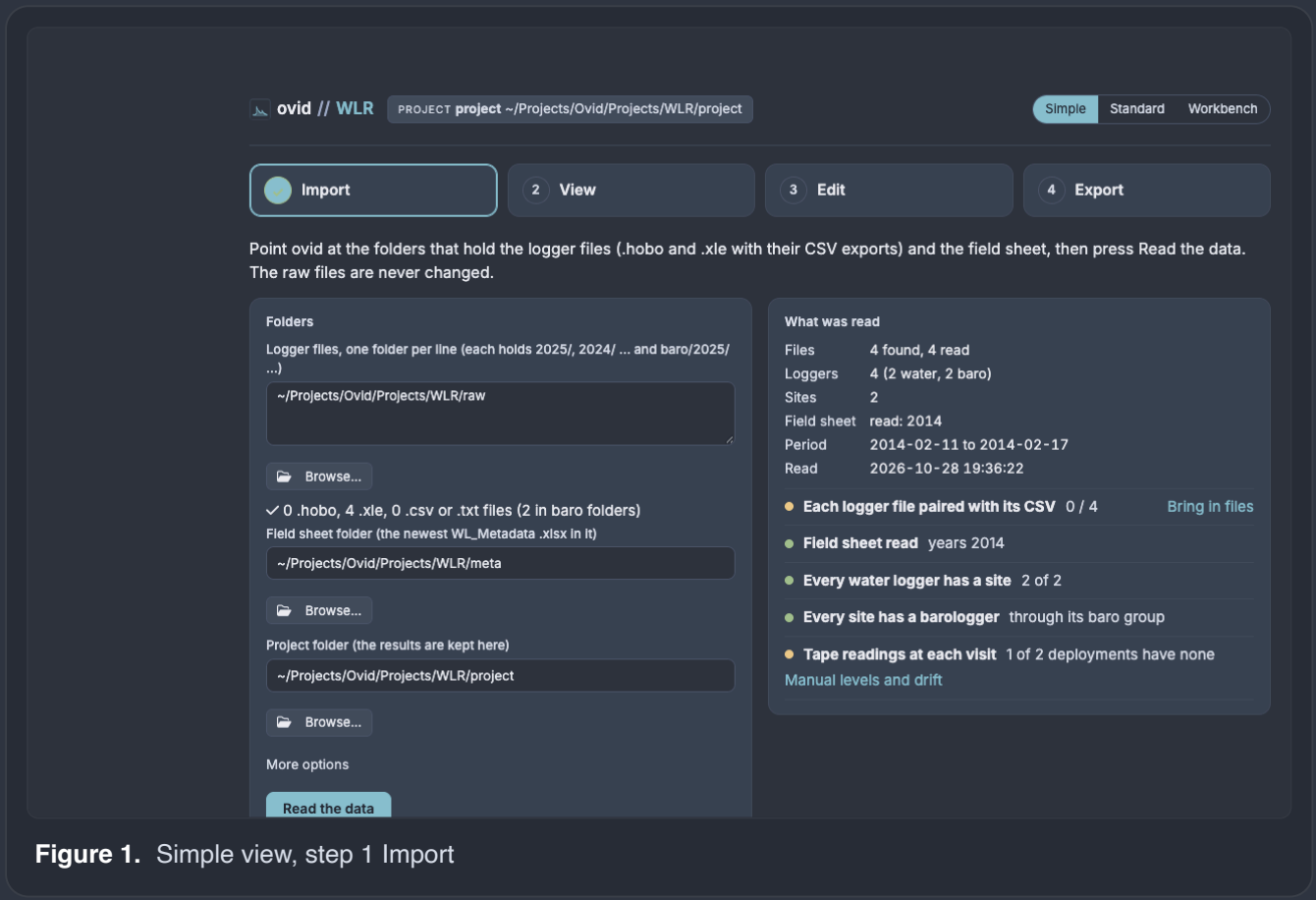


Figure 1. Simple view, step 1 Import

- **Simple**: a field check in four steps (Import, View, Edit, Export). Edit renames sites, places them, sets their land cover and moves loggers between them; it changes no data value.
- **Standard**: the set-up flow these tasks use: bring in files, organize loggers and sites, correct, report.
- **Workbench**: every change to the data by hand. Removals, out-of-water periods, quality rules, clock corrections, barologger checks and corrections over a period are edited here; Standard shows those tables and points to the Workbench.

1 Add a new download

WHEN

After a field visit, with new .hobo or .xle files.

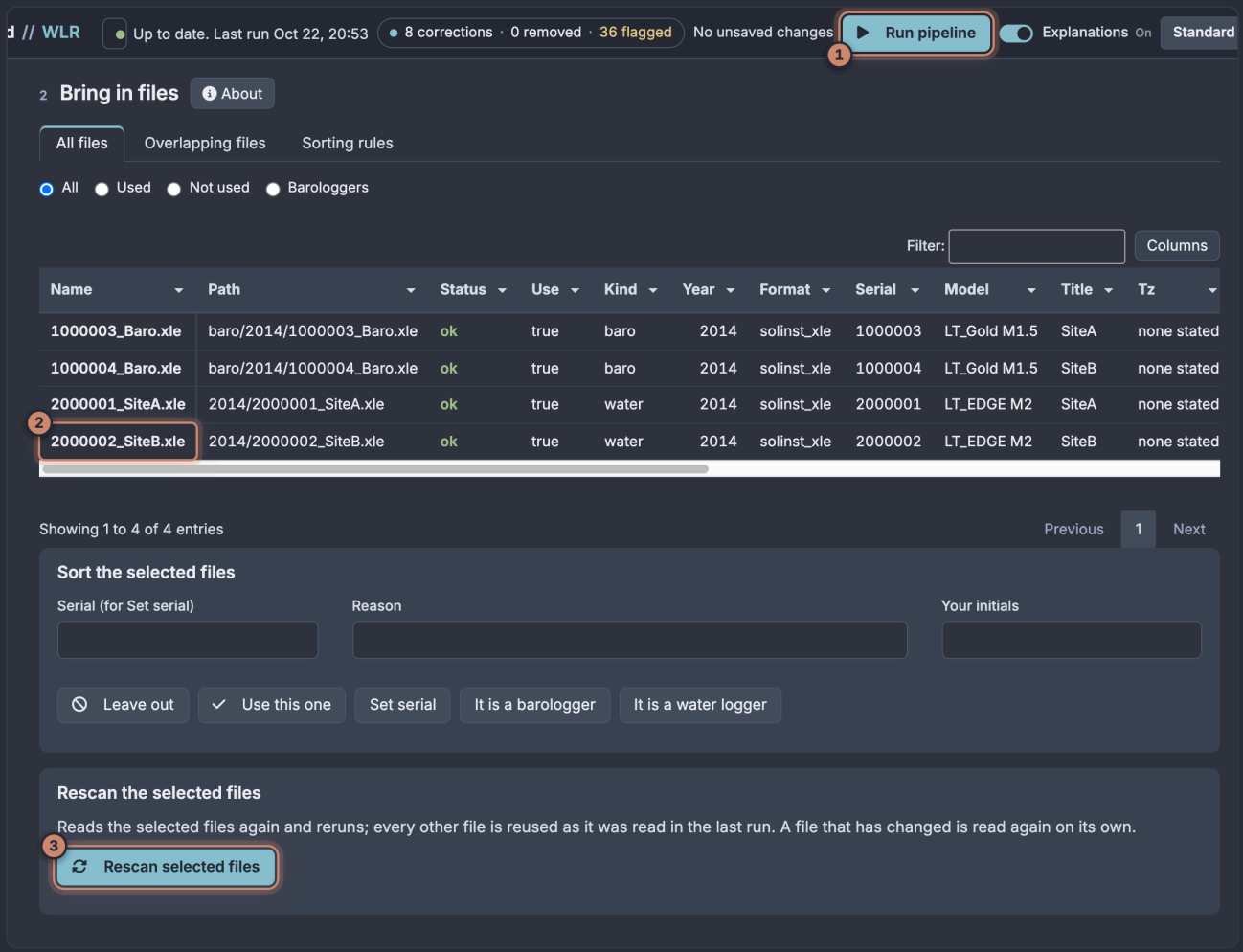


Figure 2. Run pipeline and Bring in files

1 Copy the files into the year folder of a data folder (`2025/` , or `baro/2025/` for barologgers). U20 and U20L .hobo files are read as they are; for other HOBO models, export the CSV from HOBOware beside the .hobo.

2 Press **Run pipeline** 1 .

RESULT

Only loggers whose files or set-up changed are read and drawn again; the rest are reused from the last run. A logger's downloads from several years are joined into one record, then split into deployments.

CHECK

2 Bring in files lists each file with its status. To read some files again without a full run, select them 2 and press **Rescan selected files** 3.

2 Leave out a file or say what logger it is

WHEN

A file is a bad copy, has the wrong serial in it, or is a barologger outside a `baro/` folder (or the reverse).

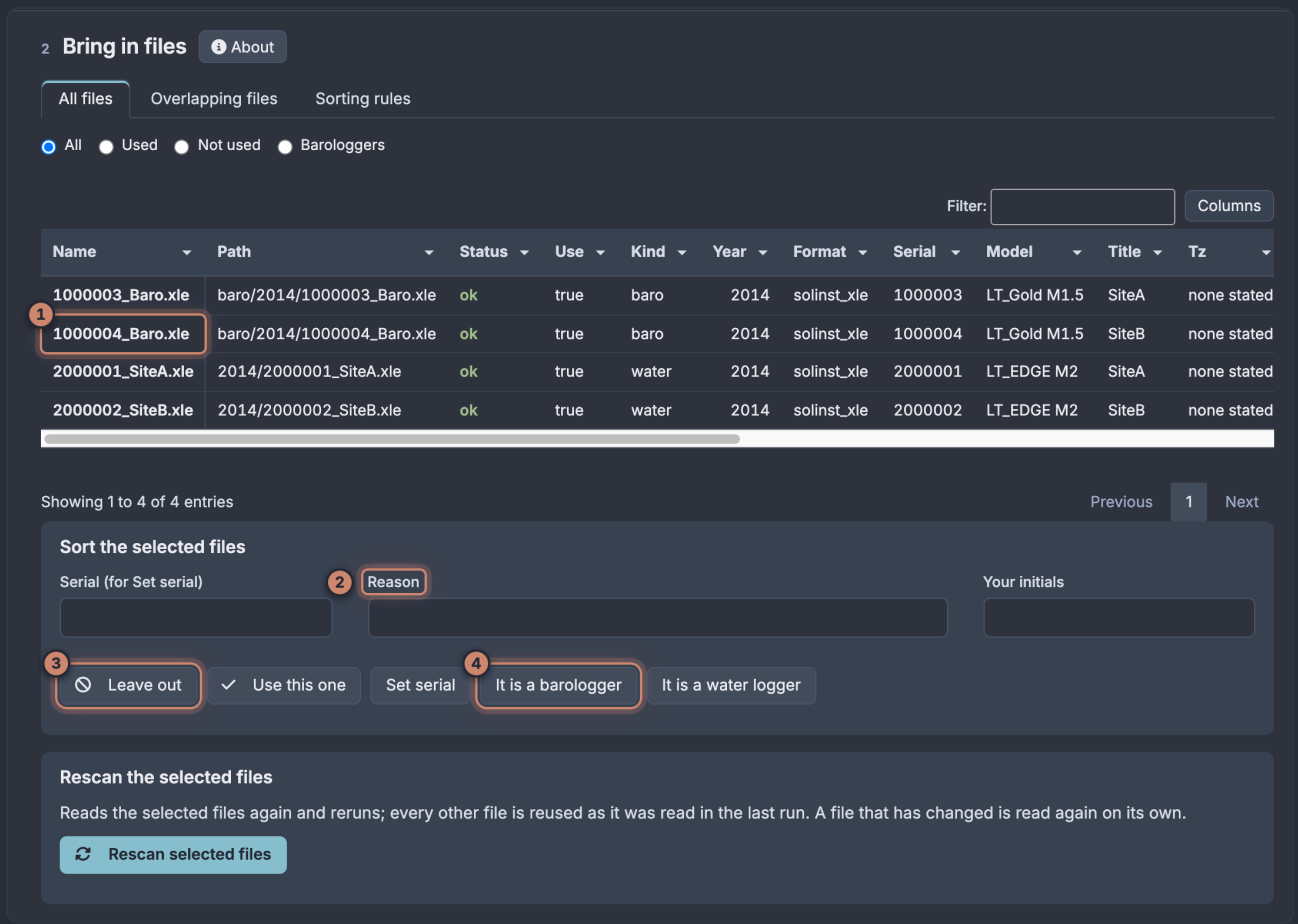


Figure 3. Bring in files: sort the selected files

1. **2 Bring in files:** select the file 1.
2. Give a **Reason** and your initials 2.
3. Press **Leave out** 3, **Set serial** (with the serial typed beside it), or **It is a barologger** / **It is a water logger** 4.

4 Press **Run pipeline**.

RESULT

A sorting rule is added to the set-up tables and applied on every run. The raw file is not changed.

CHECK

The file's **Use** and **Kind** columns show the change, and the **Audit trail** gives the reason.

3 Give a logger a site, dates or a hang length

WHEN

A logger has no site ("Unlisted" in the outputs), was moved, or the workbook is missing.

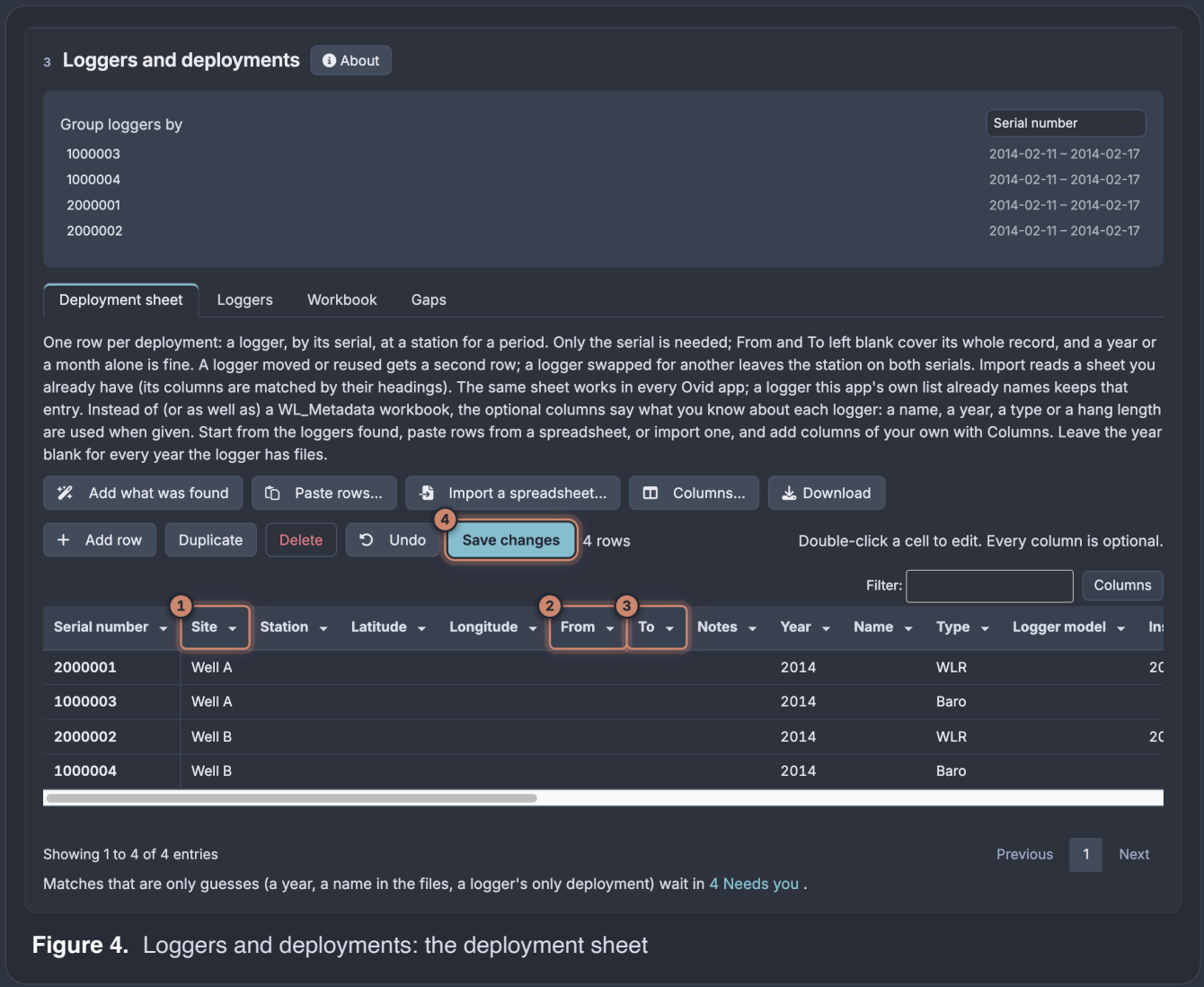


Figure 4. Loggers and deployments: the deployment sheet

1. **3 Loggers and deployments:** double-click a cell of the logger's row to edit it. **Site** 1 is the one column most rows need.
2. **From** 2 and **To** 3 limit the row to a period; blank covers the logger's whole record, and a year or a month alone is fine. A logger moved to another site gets a second row.
3. **Hang length (cm)**, the T-coupler and casing columns and the others (scroll right) are used when given.
4. **Save changes** 4, then **Run pipeline**.

RESULT

The logger's records go to that site for that period. What you type here wins over the workbook, which wins over what the files say.

CHECK

Deployment history lists every deployment with its start, end and where each came from.

4 Accept suggestions and decide what the run found

WHEN

After a run, when **Data check** in the side panel shows something to look at.

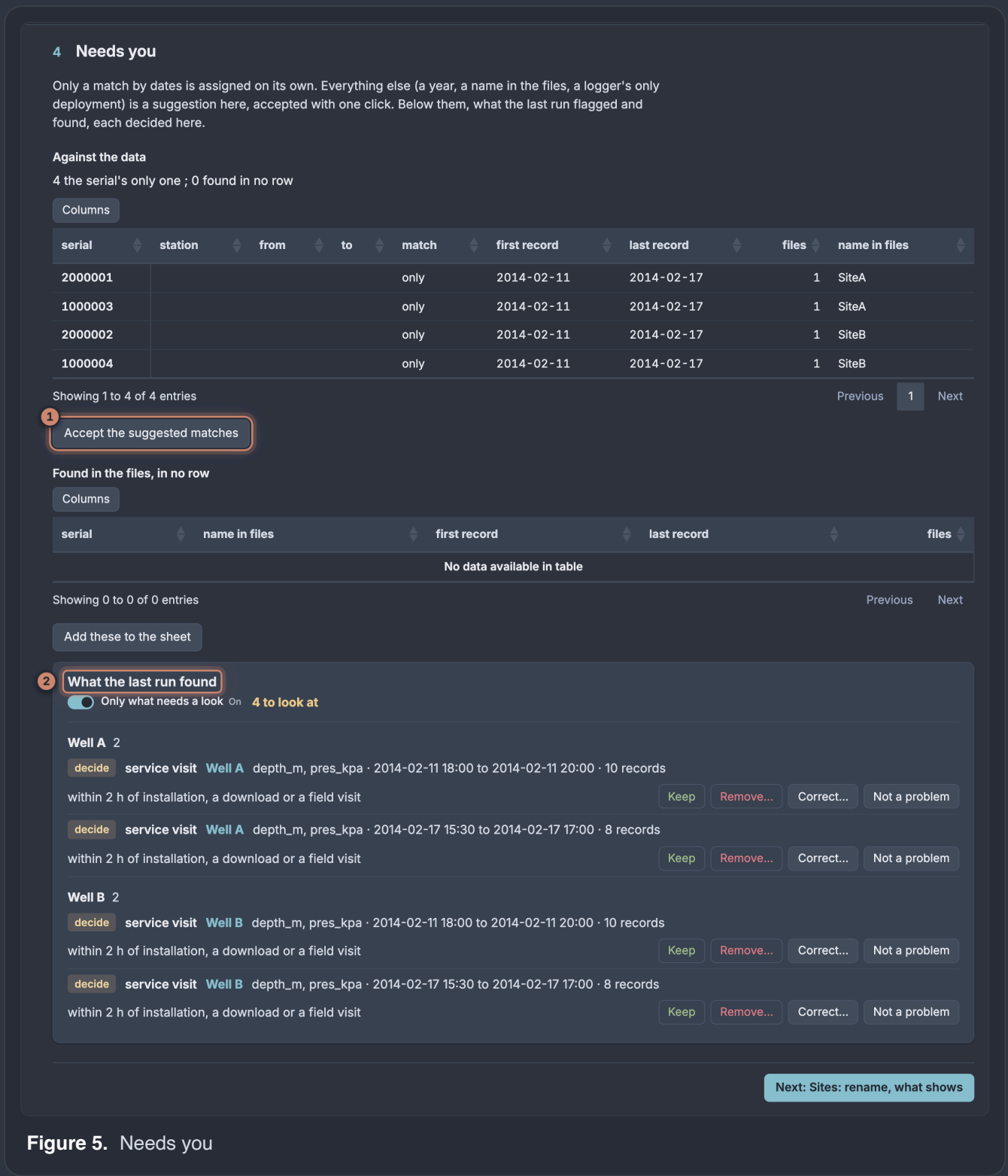


Figure 5. Needs you

1. **4 Needs you:** **Accept the suggested matches** (1) of loggers to rows, made from a year, a name in the files or a logger's only deployment.
2. Under **What the last run found** (2), decide each line: **Keep**, **Remove…**, **Correct…** or **Not a problem**, or follow its link to the page that fixes it.
3. Each decision is saved as you make it (**Not a problem** asks why); the next run uses it.

> **RESULT**
> A kept flag stays in the data as a flag. A removal or correction becomes a row in Corrections over a period, noted "Needs you".

> **CHECK**
> The item leaves the list. Turn **Only what needs a look** off to see decided items with their decision.

5 Rename a site or hide it

WHEN

A site was named wrongly, or a site should stay out of the overview pages.

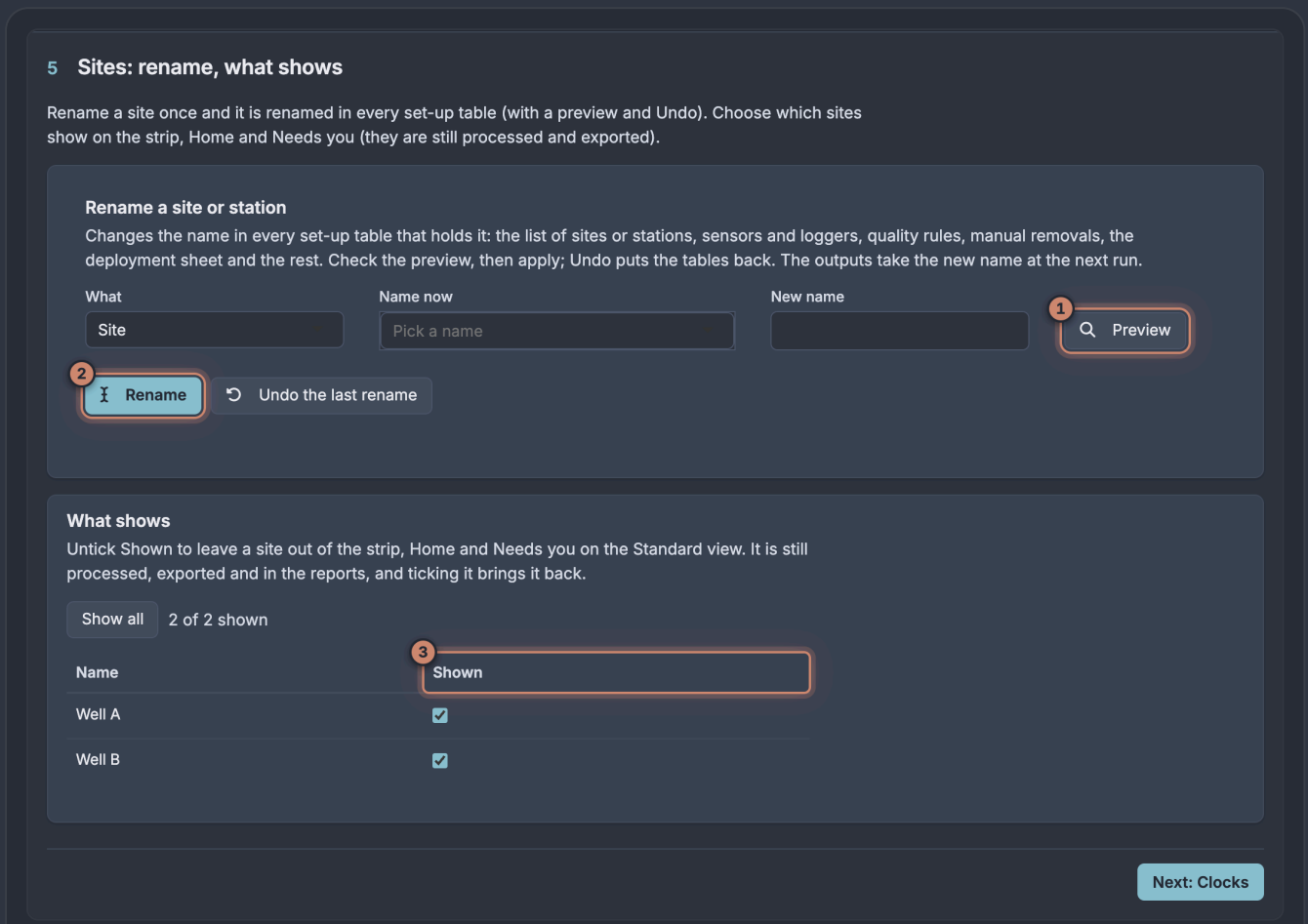


Figure 6. Sites: rename, what shows

1. **5 Sites: rename, what shows:** pick the site, type the new name and press **Preview** 1, then **Rename** 2. Every set-up table that holds the name is changed; **Undo the last rename** puts them back.
2. Untick **Shown** 3 to leave a site out of the site strip, Overview and Needs you. It is still processed, exported and in the reports.

RESULT

The outputs take the new name at the next run.

6 Correct the workbook for one deployment

WHEN

The WL_Metadata workbook has a wrong hang length, date or site for one logger, and the workbook itself should stay as it is.

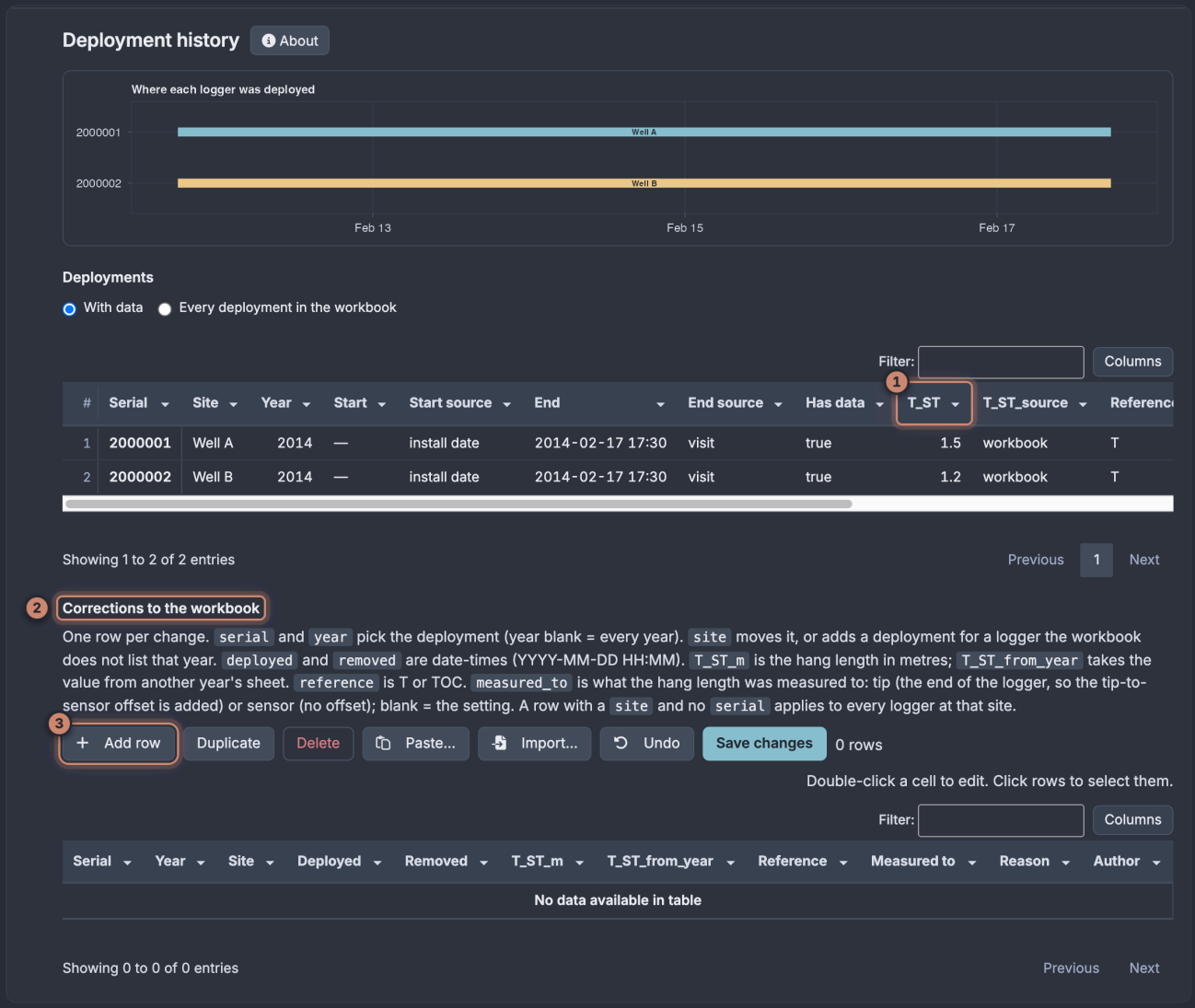


Figure 7. Deployment history

1. **Deployment history:** check the deployment's **T_ST** (1), its start and end, and where each value came from.
2. Under **Corrections to the workbook** (2), **Add row** (3): serial and year pick the deployment; fill in only what is wrong (site, deployed, removed, the hang length `T_ST_m` in metres, or `T_ST_from_year` to take another year's value), with a reason and your initials.
3. **Save changes**, then **Run pipeline**.

RESULT

The correction wins over the workbook for that deployment; the workbook is not changed.

CHECK

Set-up check (More) lists every deployment's hang length and its source.

7 Set the time zone of files that state none

WHEN

Some files that state no time zone (Solinst files, or CSVs exported without one) were logged in a different zone from the one given at the first run.

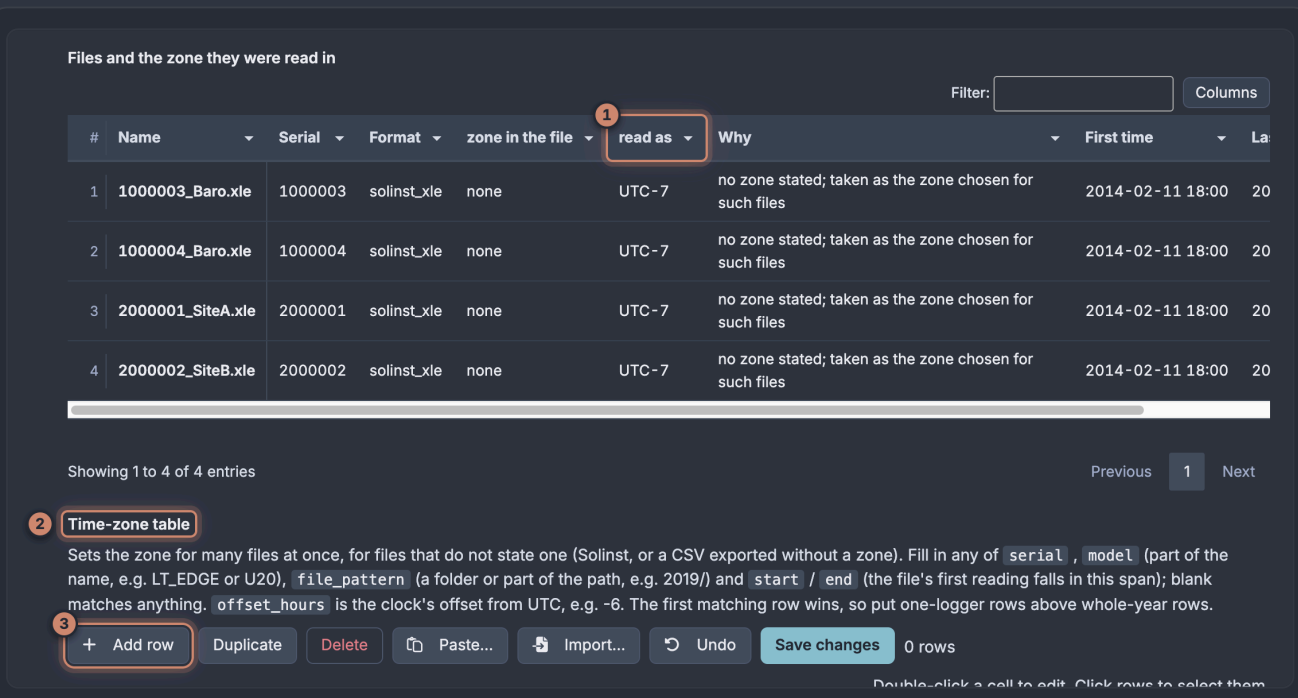


Figure 8. Clocks: files and the time-zone table

1 **Clocks:** **read as** 1 shows the zone each file was read in, and why.

2 In the **Time-zone table** 2, **Add row** 3 and fill in any of serial, model, file pattern (a folder such as `2019/`), and start and end; `offset_hours` is the clock's offset from UTC, e.g. **–6**.

3 **Save changes**, then **Run pipeline**. The first matching row wins, so put one-logger rows above whole-year rows.

RESULT

Those files are read in their own zone, then moved to the project's time zone (Settings) like every other record.

CHECK

The **read as** column shows the new zone, and the site plot has no whole-hour step at the download.

8 Fix a logger clock

WHEN

A logger's clock was wrong by a known amount, or drifted and was found out at download.

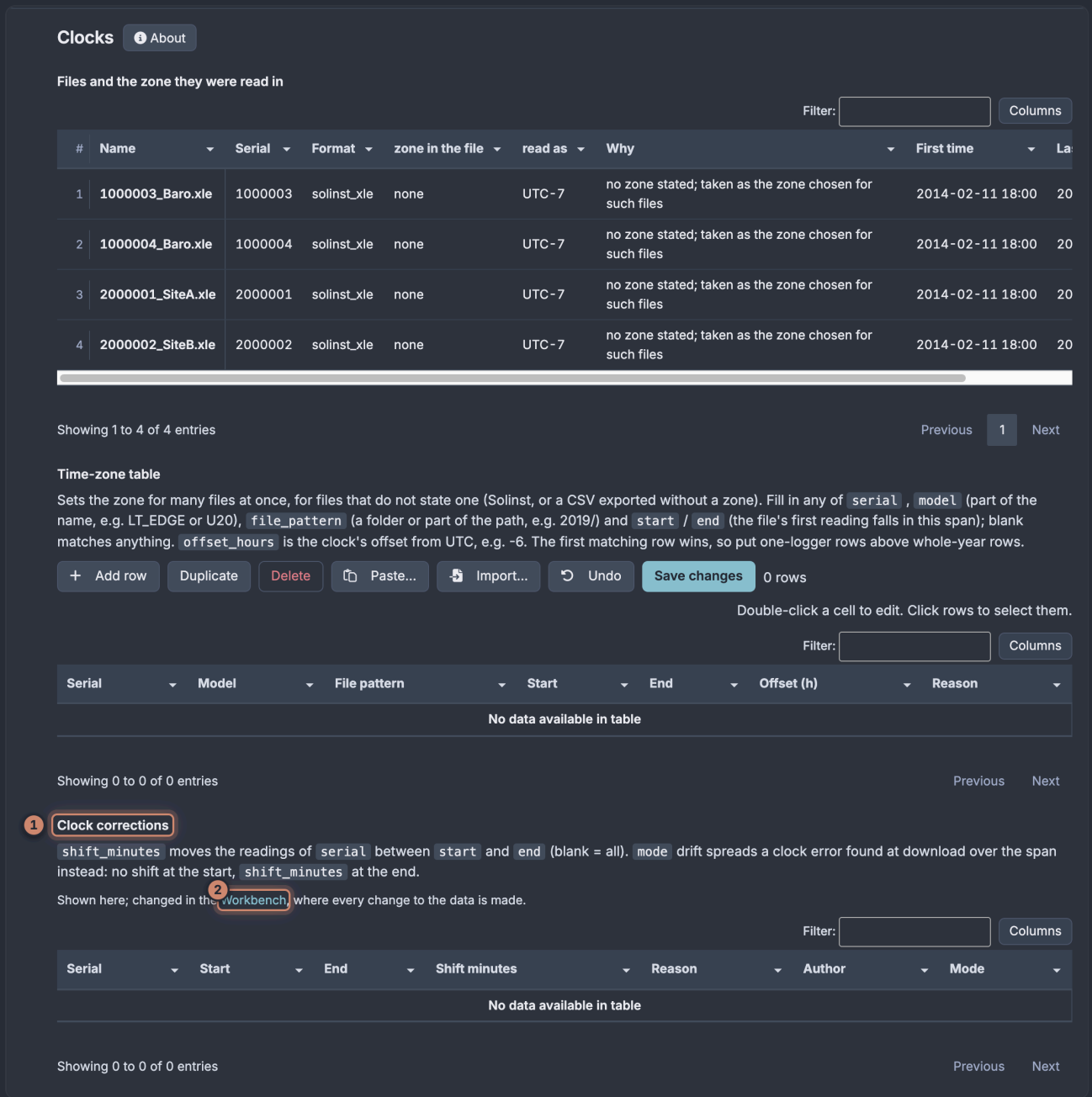


Figure 9. Clocks: clock corrections

1 On **Clocks**, **Clock corrections** (1) is shown, not edited: clock corrections change data values, so they are edited by hand in the **Workbench** (2, or the switch at the top right). There, open the **Clock corrections** table and press **Add row**.

2 Fill in the serial, the start and end (blank = all of its record) and `shift_minutes`. With `mode` set to drift, the shift grows from none at the start to `shift_minutes` at the end.

3 **Save changes**, then **Run**.

> **RESULT**
> The readings are moved in time; each shifted record is in the audit trail.

> **CHECK**
> On the site plot, events and field visits line up with the logger's record again.

9 Choose the barologgers

WHEN

A site has more than one barologger near it, a barologger failed, or the suggested one is not the best.

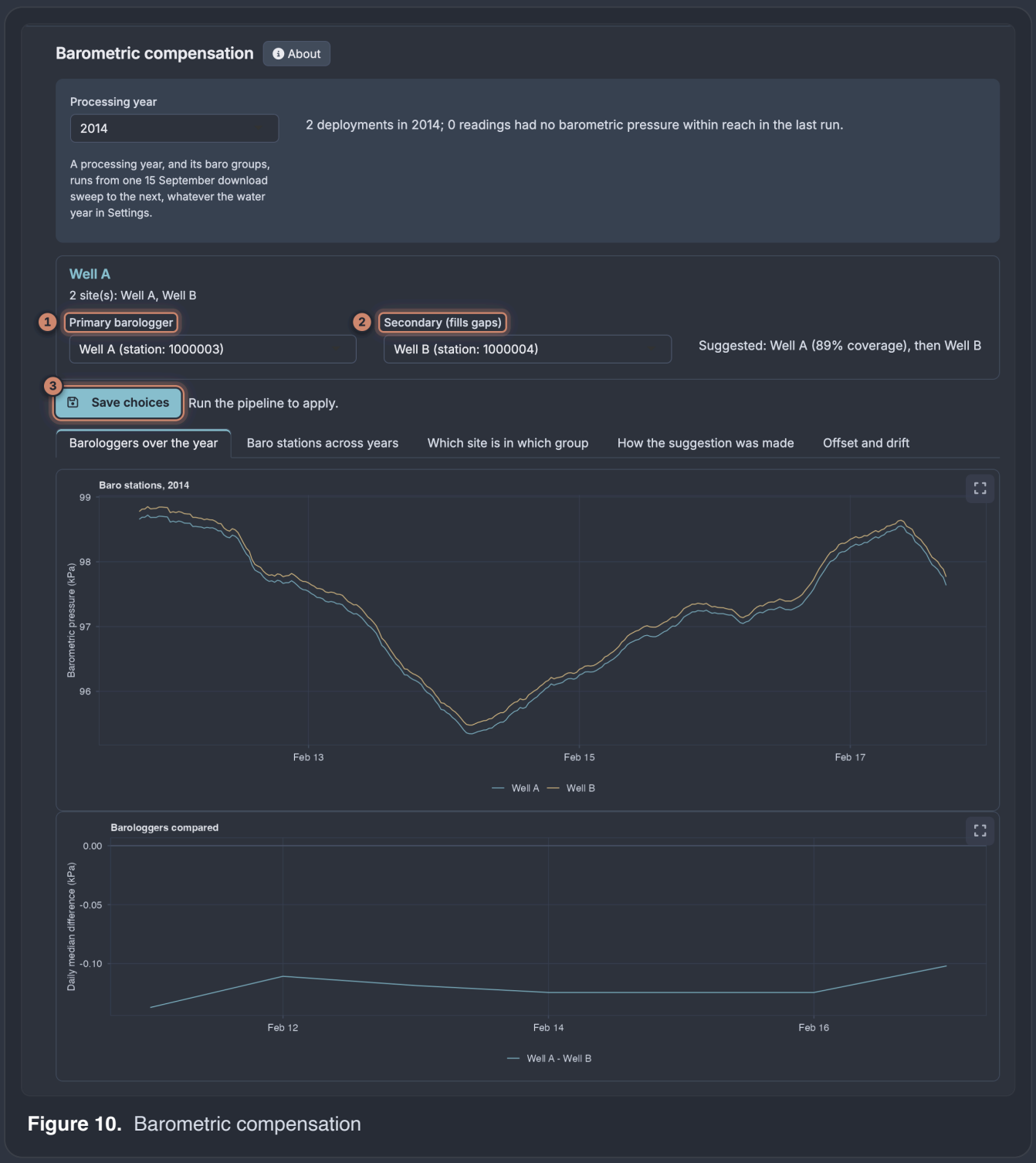


Figure 10. Barometric compensation

1. **Barometric compensation:** pick the **Processing year**, then for each group the **Primary barologger** 1 and a **Secondary** 2 that fills its gaps.
2. **Save choices** 3, then **Run pipeline**.

> **RESULT**
> Every water reading is compensated with the primary's air pressure at its time, or the secondary's where the primary has none; a fill-in from another barologger is shifted onto the primary's level and flagged.

> **CHECK**
> **Barologgers over the year** plots each candidate; **Offset and drift** shows how far they differ.

10 Enter a logger offset or a survey point

WHEN

Hang lengths are measured to the logger's tip, or a site has a surveyed reference point, so depths can become water elevations.



Figure 11. Datum and offsets: logger offsets

1. **Datum and offsets:** the formulas at the top show how each value is made from the hang length (T_ST), the offset and the reference elevation.

2. **Logger offsets** 1: **Set the offset for** a logger model or one logger 2, pick a known offset or **Manual measurement** and type the tip-to-sensor distance in cm 3, then **Set offset** 4.

3. For elevations: **Survey points for each site** links a site to a dGPS point (code T, TOC or GS); **Elevations typed by hand** takes a reference elevation surveyed another way. Record the vertical datum and geoid model with every elevation (the table has columns for them): elevations on different datums cannot be compared.

4. Press **Run pipeline**.

> **RESULT**
> Every deployment with a hang length gets depth to water below its reference point; those with a surveyed reference also get sensor and water elevations.

> **CHECK**
> **Deployment history** shows the offset used for each deployment and where it came from.

11 Enter tape checks and check drift

WHEN

After each field visit with a tape reading of the water level.

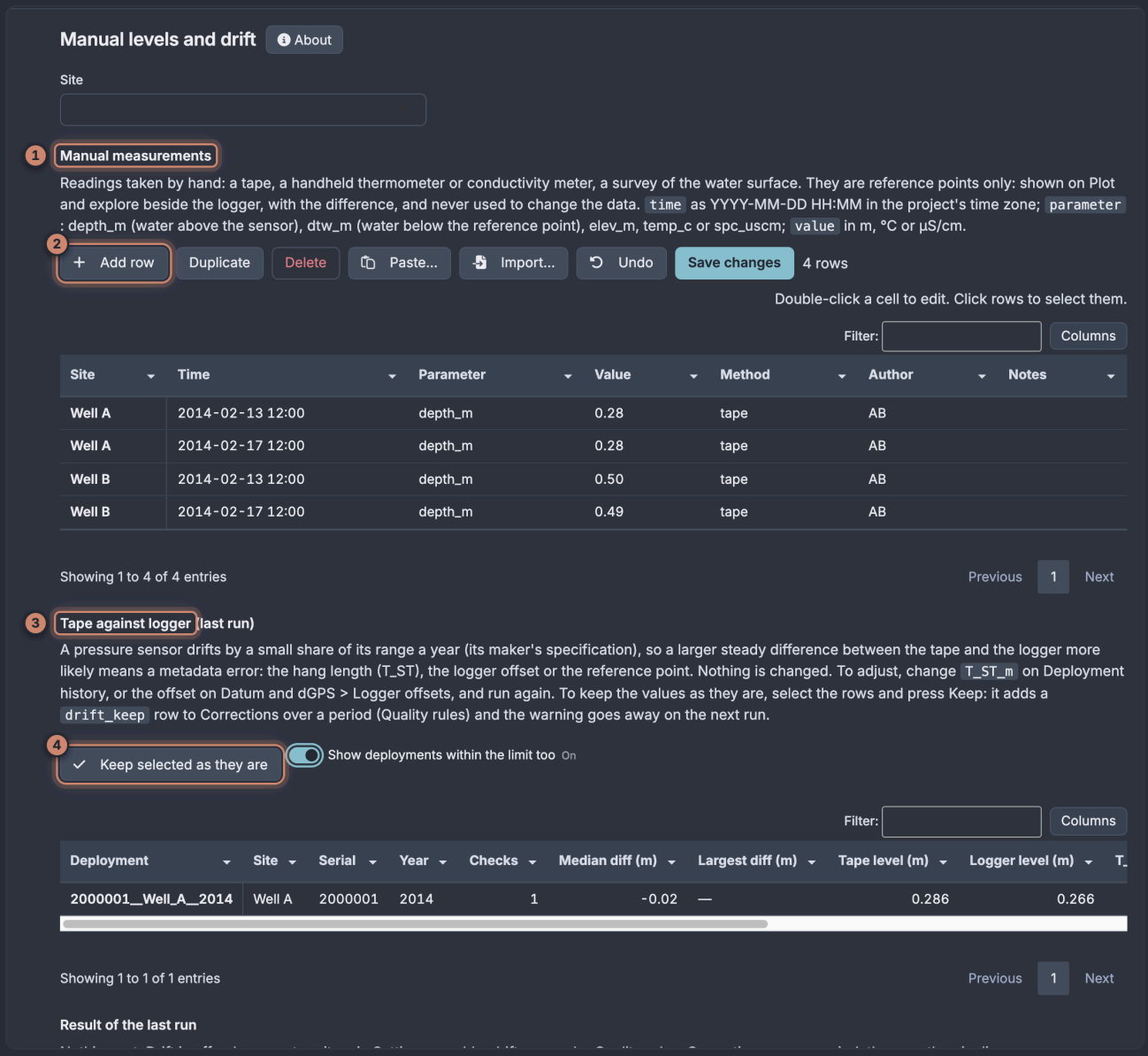


Figure 12. Manual levels and drift

1 The tape reading at each download goes with the deployment: **T-coupler to water (cm)** (or casing to water) in the WL_Metadata workbook or on the deployment sheet (task 3).

2 Other readings by hand go on **Manual levels and drift**: under **Manual measurements** 1, **Add row** 2 with the site, time (YYYY-MM-DD HH:MM in the project's zone), parameter (`depth_m` water above the sensor, `dtw_m` depth to water, `elev_m` water elevation, `temp_c` or `spc_uscm`) and value, then **Save changes**. They show on the plots and on the site's **Manual vs logger** card, with the difference, and never change the data.

3 Press **Run pipeline**. **Tape against logger** 3 compares each visit's tape with the logger just before the visit.

4 A deployment with one check shows its median only. A difference beyond the limit (**5 cm** unless set in Settings) is more likely a metadata error than sensor drift: fix the hang length (task 6) or the offset (task 10). To keep the values as they are, select the rows and press **Keep selected as they are** 4.

> **RESULT**
> The readings by hand show beside the logger with their difference; the data are not changed. Drift correction from the tapes is off by default: switch it on in Settings to export corrected values beside the logger's own.

> **CHECK**
> After the run, look at the site's **Manual vs logger** card and the **Tape against logger** table.

12 Remove a stretch of data

WHEN

The logger was out of the water, frozen, disturbed during a visit, or still settling after it went into the well. The automatic check flags readings within **2 h** of installation, a download or a visit; remove those that are not water level, and anything else the check did not catch.

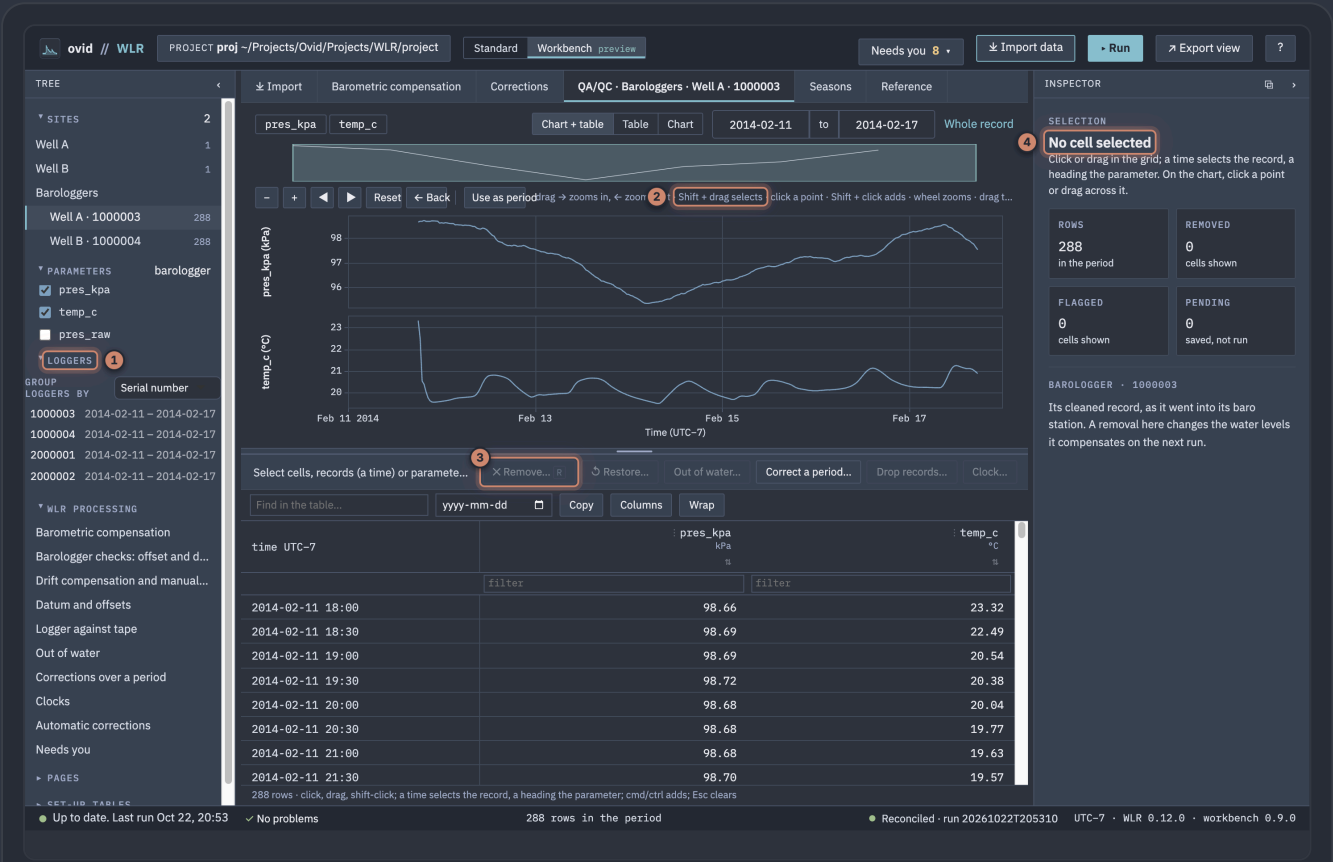


Figure 13. Workbench

1. Switch to **Workbench** (top right). In the tree, pick the site, deployment or logger 1.
2. Shift + drag across the chart 2 to select the stretch; the inspector 4 shows its count, mean, minimum, maximum and how many are already removed or flagged.
3. Press **Remove…** 3 and give a reason and your initials. **Out of water…**, **Correct a period…** and **Drop records…** beside it work the same way.
4. Press **Run**.

RESULT

A row is added to Corrections over a period. The old values stay in the audit trail and raw files are never changed. **Restore** undoes it the same way.

CHECK

After the run, the stretch shows as removed on the plot (a faint line keeps the old values) and the **Audit trail** lists it.

13 Change a quality limit

WHEN

A limit removes good readings, or lets bad ones through, for your loggers or sites.

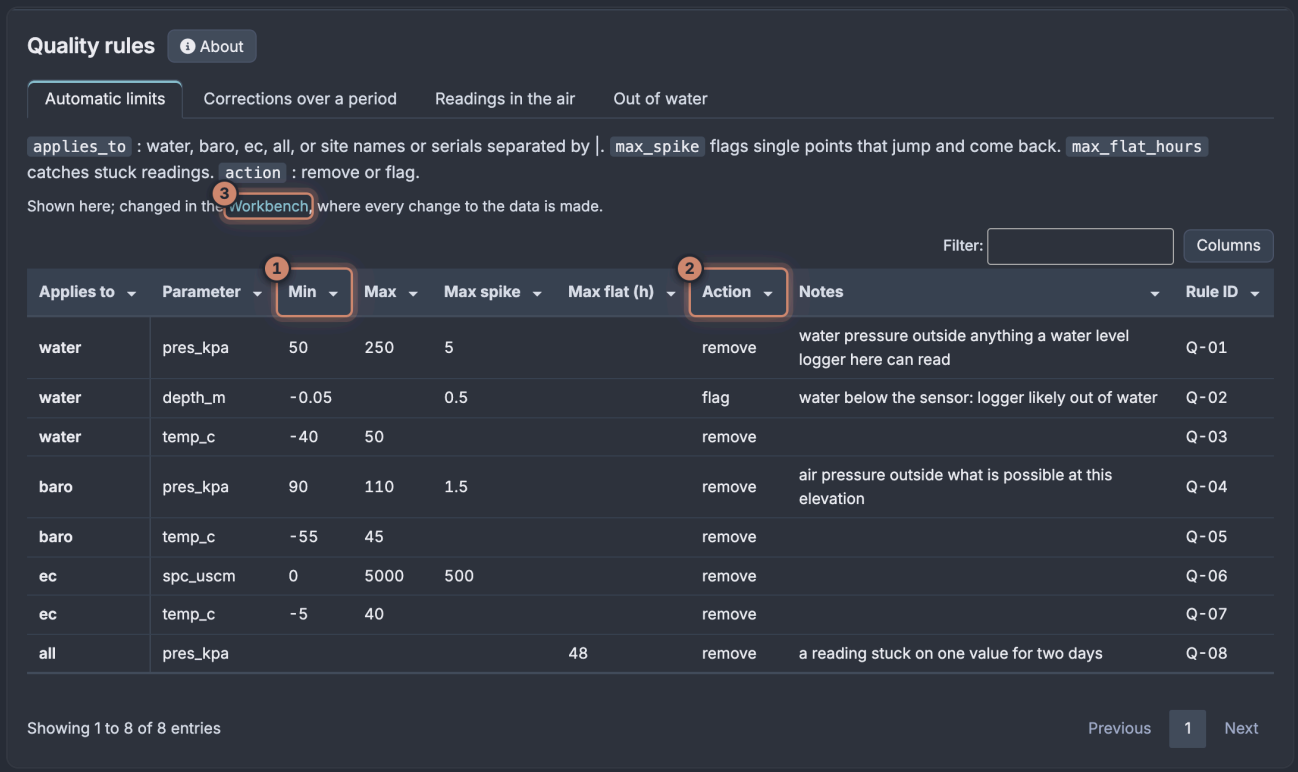


Figure 14. Quality rules

1 **Quality rules** shows the limits; they are edited by hand in the **Workbench** (3, or the switch at the top right). There, open the **Quality rules** table and double-click a cell to change it: **Min** and **Max** (1), the largest single jump (max spike) or the longest flat stretch (max flat h).

2 **Action** 2: remove or flag. **Applies to** takes water, baro, ec, all, or site names or serials separated by |.

3 **Save changes**, then **Run**.

RESULT

The rule is applied to every reading it covers; each removal or flag carries the rule's ID.

CHECK

The **Audit trail** lists the rule's ID and how many values it removed or flagged.

NOTE

The seeded limits are ovid defaults with no published source (reference guide, section 13); check them against your loggers' specifications. The barometric range, **90** to **110 kPa**, fits sites from sea level to about **1,000 m**: widen Q-04 for higher sites, or good readings are removed.

14 Switch an automatic correction off

WHEN

A correction the run makes on its own does not suit the project.



6 Automatic corrections

Corrections every run makes on its own, each with a switch. They are on by default.

Everything a run changes in the data on its own, without a row in a set-up table asking for it, and where each is switched. Flags never change a value. A switch here is saved with the app's settings and used from the next run.

Kind	What the run does	1 State	2 Where to change it
Removes values	Records with corrupt time stamps among good ones (off the time grid and out of order) are left out; each is in the audit trail.	On	
Removes values	Records stamped before the first year or in the future, or after the clock jumped back by more than a day within a file, are left out as a reset clock (off: kept at the time they carry; records with no time are always left out).	On	
Changes values	Every time stamp is moved to the base zone (the time zone in Settings).	always	Settings (base zone)
Changes values	Logger clocks are shifted by the rows of the clock corrections table.	by table, none	Clock corrections
Changes values	The water a Solinst Gold or Junior adds for the altitude it was programmed with is taken off again.	on	Settings, Units and conversion
Fills in values	Barometric readings are interpolated linearly across gaps up to the hours under Settings, Advanced (off: only water readings at a barometric reading's time are compensated).	On	
Fills in values	Where neither chosen barologger has a reading, another one fills in (shifted onto the primary's level, flagged).	on	Settings, Checks
Changes values	Barologger readings are corrected by the rows of the baro corrections table.	by table, none	Barometric compensation
Changes values	The tip-to-sensor distance is added where the hang length was measured to the tip.	off	Settings, Units and conversion
Changes values	Water levels are corrected to the tape checks between visits.	off	Settings, Checks (drift between tape checks)
Removes values	Readings taken in the air at the start or end of a deployment are removed (when off they are flagged and listed as suggestions).	off	Settings, Checks
Removes values	Values outside each rule's limits are removed or flagged, as each row says.	by table, 8 rows	Quality rules
Changes values	Values in each period of the corrections list are removed (with the limits), offset, scaled, drift-corrected, filled by interpolation or flagged: pressure and temperature before barometric compensation, depth before the elevations are worked out.	by table, none	Corrections over a period
Removes values	Periods the out-of-water table lists are removed.	by table, none	Out of water
Flags values (none changed)	Flags decided not a problem (Needs you) are left out of the exported flag columns; flags.csv keeps them, marked accepted.	On	
Flags values (none changed)	Readings close to installation, a download or a field visit, with too little water above the sensor, or well below the air pressure are flagged.	always	Settings, Checks (service window, out of water)

Next: Report

Figure 15. Automatic corrections

1. **6 Automatic corrections** lists everything a run changes on its own, with its **State** (1) and **Where to change it** (2).
2. Turn its switch off, or change the setting named in the last column, then **Run pipeline**.

> **RESULT**
> From the next run the correction is not made; flags never change a value either way.

> **CHECK**
> The **State** column shows the change; after the run, the **Audit trail** no longer lists that correction.

15 Look at a site over time

WHEN

To look at a deployment in detail, compare its values, or save a figure or the data in view.

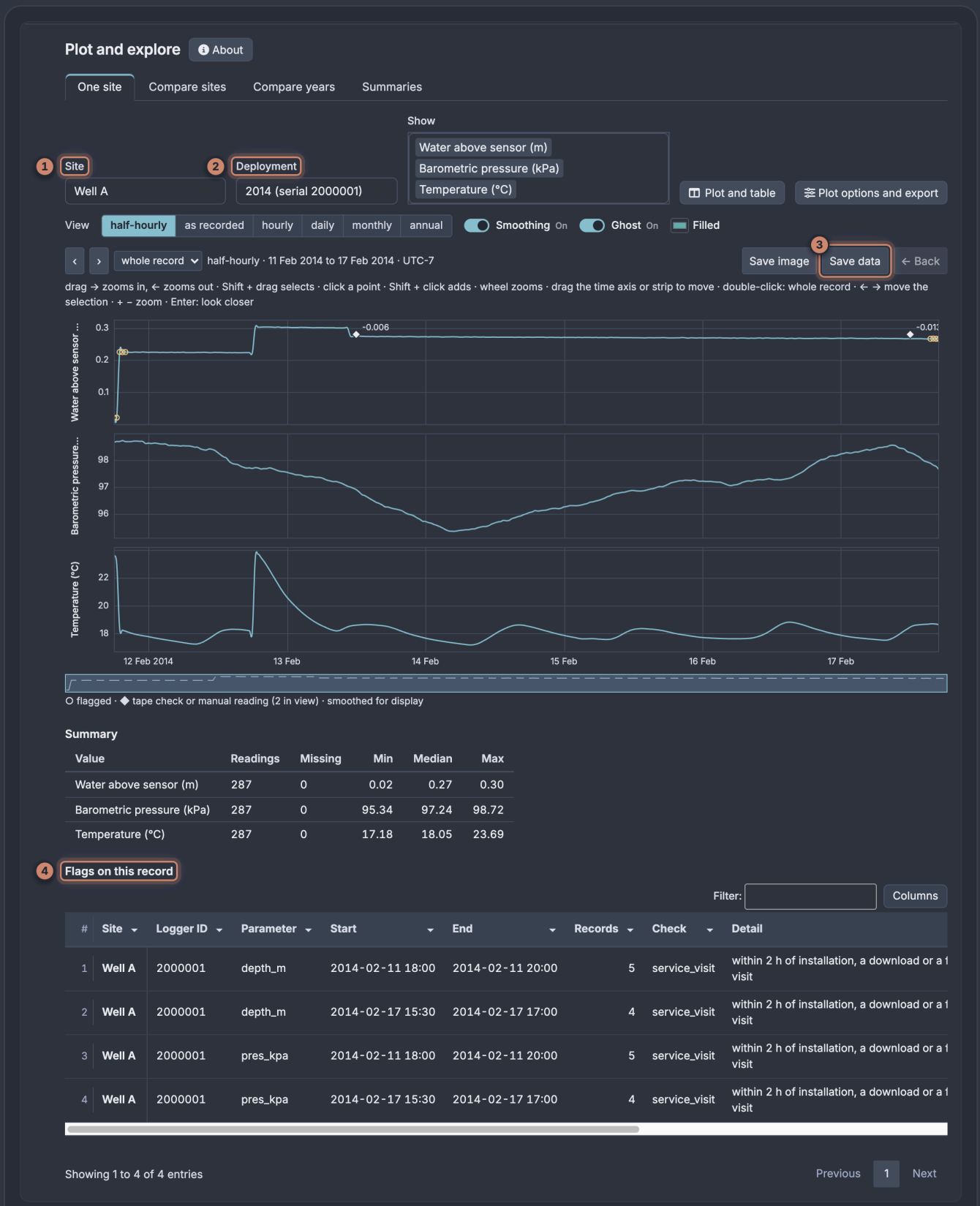


Figure 16. Plot and explore

1 **Plot and explore** (More): pick the **Site** 1 and **Deployment** 2, and the values to show.

2 Drag left to right zooms in, right to left zooms out; Shift + drag selects a stretch (to remove it, use the Workbench, task 12); double-click shows the whole record. The line is smoothed for display (**Smoothing** in the plot options turns it off); **Save data** gives the readings themselves.

3 **Save image** or **Save data** 3 saves what is in view.

4 **Flags on this record** 4 lists every flag with its check and rule.

> **RESULT**
> Nothing changes in the project; the plot and the data saved are what is in view.

16 Find out why a value was removed

WHEN

A value is missing from an export or a plot and you want to know why.

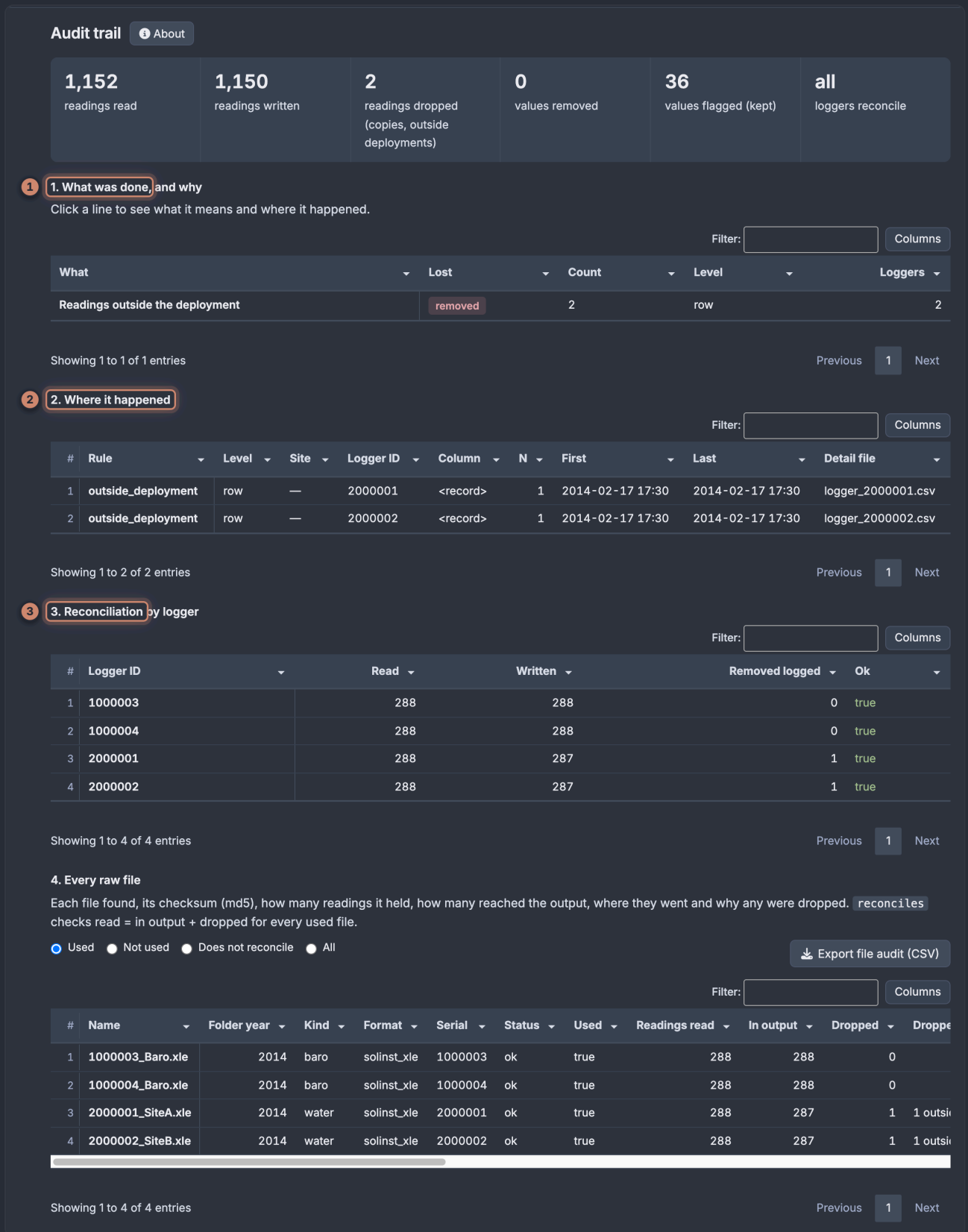


Figure 17. Audit trail

1 **Audit trail** (More): read it top down: what was done and why 1, where it happened 2, then each logger's reconciliation 3 and every raw file.

2 Click a line in 1 to see what it means and where it happened.

RESULT

Every reading read is either written or accounted for, logger by logger.

17 Run the set-up check

WHEN

Before the first run, after changing the set-up, and before exporting.

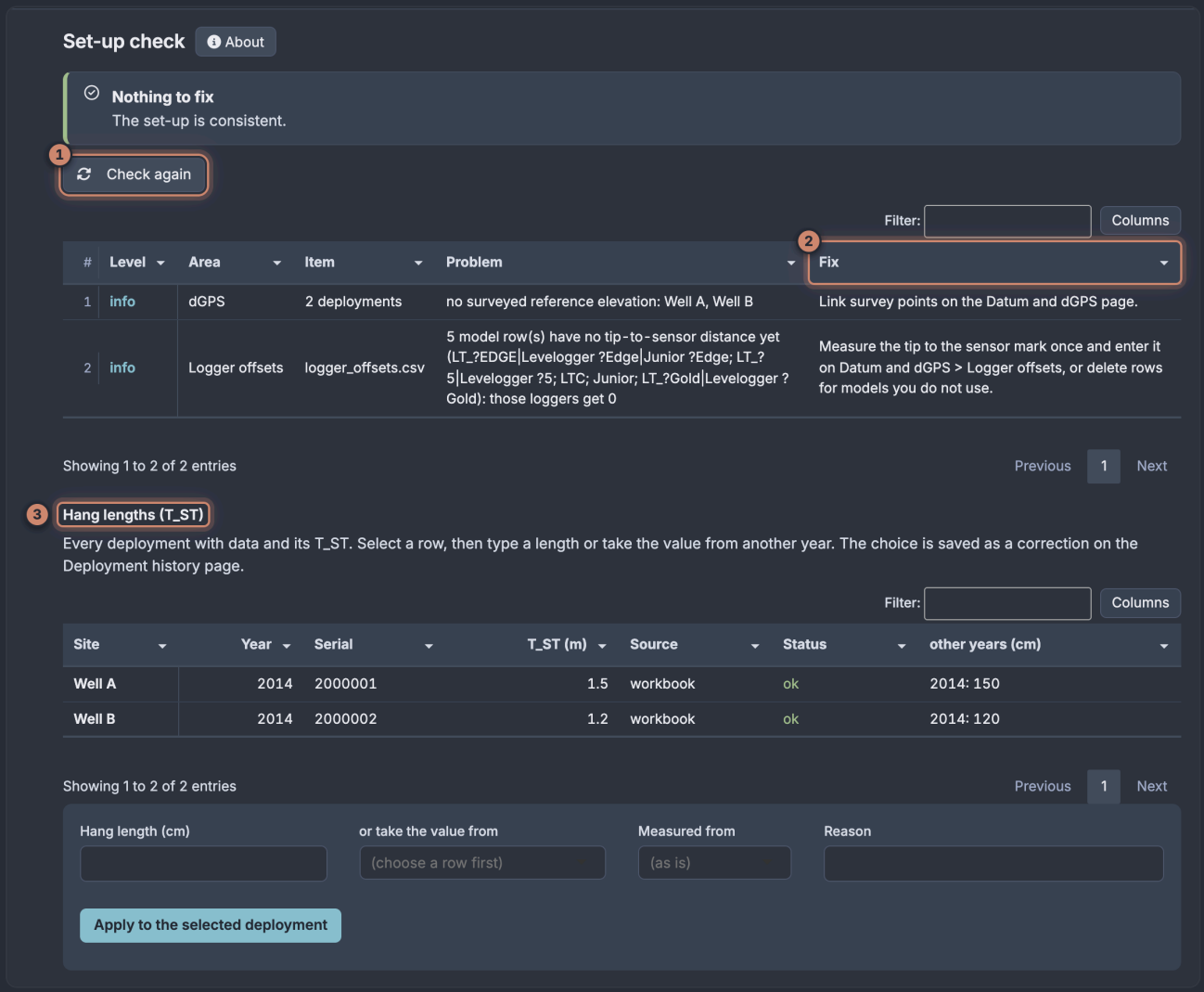


Figure 18. Set-up check

1 **Set-up check** (More) lists what is missing or inconsistent in the set-up, each with a **Fix** 2. Fix what is red first.

2. **Check again** 1 after a change.
3. **Hang lengths (T_ST)** 3: every deployment with data and its hang length; select a row and type a length or take another year's value.

> **RESULT**
> Nothing changes until you follow a fix and run again.

18 Export data and reports

> **WHEN**
> To send data to someone, archive it, or build reports.

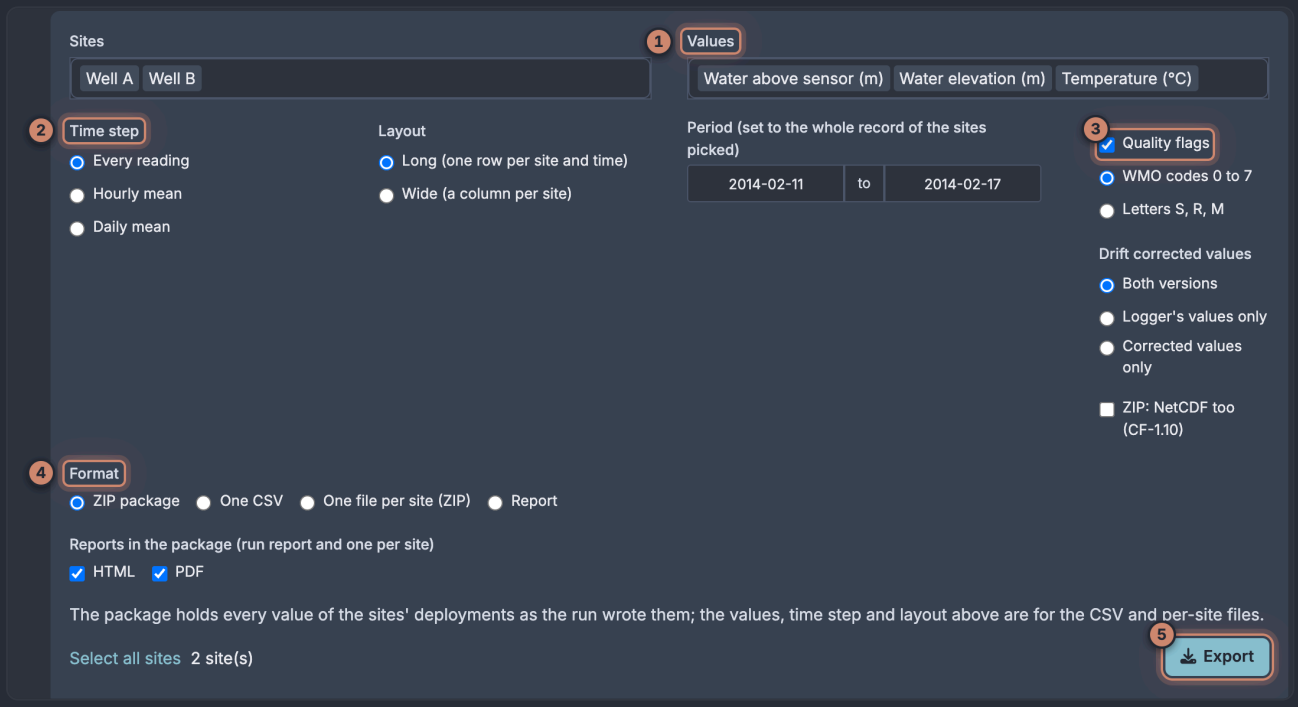


Figure 19. Report

1. **7 Report:** pick the sites, the **Values** 1 and the **Time step** 2.
2. **Quality flags** 3: WMO codes 0 to 7, or letters S, R, M. Drift-corrected values can be included beside the logger's own.
3. **Format** 4: a ZIP package (NetCDF too if ticked), one CSV, one file per site, or the report alone. Reports in HTML, PDF or both.
4. **Export** 5.

RESULT

The package holds every value of the sites' deployments as the run wrote them, the manual measurements beside the logger, the reports and a README with the processing level of each file; it can be used without the app. Fill in the **Dataset** card in Settings first: it is written into every NetCDF file.

19 Back up or share the set-up

WHEN

To keep a copy of the set-up, or start another project with it.

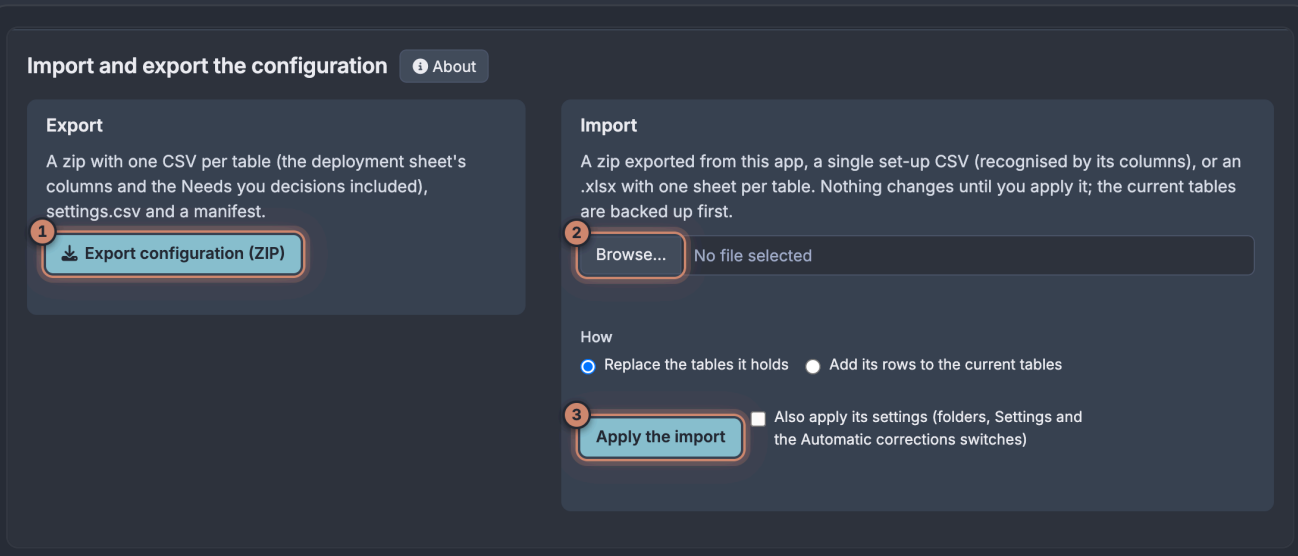


Figure 20. Configuration

1. **Configuration** (More): **Export configuration (ZIP)** 1 writes every set-up table, the deployment sheet, the Needs you decisions and the settings.
2. In another project: **Browse…** for the zip 2, choose to replace the tables or add its rows, then **Apply the import** 3.

RESULT

The tables it replaces are kept in the project's history folder (config/history).

20 Report a problem

WHEN

A page shows a notice with a reference, or something looks wrong.

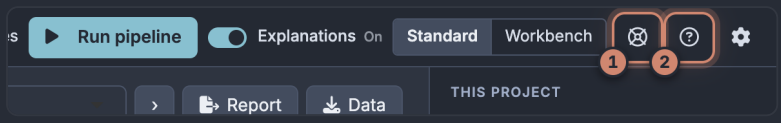


Figure 21. Top bar: troubleshooting and help

1. A notice with a reference (E- and six characters) means part of a page failed; the app carries on.
2. The lifebuoy button 1 builds a troubleshooting zip: versions, logs, problems and the set-up, no raw data. Send it with the reference.
3. The ? button 2 opens the reference guide.

RESULT

The zip and the reference let a problem be traced without your raw data.

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ovid WLR. For what WLR does to the data and why, see the WLR reference guide; every figure can be retaken with docs/guide_shots.R.